

02477 Bayesian Machine Learning Exam 2025: Solution

```
In [1]: %matplotlib inline
import jax.numpy as jnp
import numpy as np
from jax import random

import matplotlib.pyplot as plt
```

Part 1

Question 1.1

The definition of the model $y_n = f(x_n) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ implies a Gaussian likelihood of the form $p(y_n | \mathbf{w}, \phi(x_n)) = \mathcal{N}(y_n | \mathbf{w}^T \phi(x_n), \sigma^2)$:

$$p(\mathbf{y} | \mathbf{w}, \Phi) = \mathcal{N}(\mathbf{y}_n | \mathbf{w}^T \phi(x_n), \sigma^2) = \mathcal{N}(\mathbf{y} | \Phi \mathbf{w}, \sigma^2 \mathbf{I}), \quad (1)$$

where Φ is given by

$$\Phi = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}.$$

```
In [2]: x = jnp.array([0.0, 1.0, 2.0, 4.0, 5.0])
y = jnp.array([1.0, 0.5, -0.1, -0.9, 1.1])

design_matrix = lambda x: jnp.column_stack([jnp.ones(len(x)), x, x**2, x**3])
Phi = design_matrix(x)
print('Design matrix Phi\n', np.array2string(Phi, precision=2))
```

```
Design matrix Phi
[[ 1.  0.  0.  0.]
 [ 1.  1.  1.  1.]
 [ 1.  2.  4.  8.]
 [ 1.  4. 16. 64.]
 [ 1.  5. 25.125.]]
```

Question 1.2

The maximum likelihood solution for a linear model with Gaussian likelihood is given by the normal equations:

$$\mathbf{w}_{\text{MLE}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} = \begin{bmatrix} 0.98 \\ 0.27 \\ -0.69 \\ 0.13 \end{bmatrix} \quad (2)$$

```
In [3]: w_MLE = jnp.linalg.solve(Phi.T@Phi, Phi.T@y)
print('MLE for w:', np.array2string(w_MLE, precision=2))
```

MLE for w: [0.95 0.27 -0.69 0.13]

Question 1.3

The posterior distribution for a Gaussian linear model is:

$$p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3)$$

where

$$\mathbf{m} = \frac{1}{\sigma^2} \mathbf{S} \Phi^T \mathbf{y} = \begin{bmatrix} 0.95 \\ 0.25 \\ -0.68 \\ 0.13 \end{bmatrix}, \quad \mathbf{S} = \left(\alpha \mathbf{I} + \frac{1}{\sigma^2} \Phi^T \Phi \right)^{-1} = \begin{bmatrix} 0.04 & -0.04 & 0.01 & 0.00 \\ -0.04 & 0.13 & -0.06 & 0.01 \\ 0.01 & -0.06 & 0.04 & 0.00 \\ 0.00 & 0.01 & 0.00 & 0.00 \end{bmatrix} \quad (4)$$

```
In [4]: alpha = 1/2
sigma = 1/5

S_inv = alpha*jnp.identity(4) + (1/sigma**2)*Phi.T@Phi
S = jnp.linalg.inv(S_inv)
m = (1/sigma**2) * jnp.linalg.solve(S_inv, Phi.T@y)

print('Posterior m:\n', np.array2string(m, precision=2), '\n')
print('Posterior S:\n', np.array2string(S, precision=2))
```

Posterior m:
[0.95 0.25 -0.68 0.13]

Posterior S:
[[0.04 -0.04 0.01 -0.]
[-0.04 0.13 -0.06 0.01]
[0.01 -0.06 0.04 -0.]
[-0. 0.01 -0. 0.]]

Question 1.4

The posterior predictive distribution for a Bayesian linear model is given by

$$p(y^*|\mathbf{y}, x^*) = \mathcal{N}(y^*|\mathbf{m}^T \phi_*, \phi_*^T \mathbf{S} \phi_* + \sigma^2) \approx \mathcal{N}(y^*| -1.00, 0.07) \quad (5)$$

```
In [5]: x_star = 3.
phi_star = design_matrix(jnp.array([x_star])).ravel()

pred_mean = m.T @ phi_star
pred_var = phi_star.T @ S @ phi_star + sigma**2

print(f'p(y*|y, x*) = N(y*| {pred_mean:3.2f}, {pred_var:3.2f})')
```

p(y*|y, x*) = N(y*| -1.00, 0.07)

Part 2

Question 2.1

The approximate posterior mean is given by

$$\mathbb{E}[w_1|\mathbf{y}] \approx \sum_{w_1, w_2} w_1 q(w_1, w_2) = 0.5(0.09 + 0.12) + 0.6(0.12 + 0.2 + 0.12) + 0.7(0.12 + 0.23) \approx 0.61 \quad (6)$$

The approximate posterior variance is given by

$$\mathbb{V}[w_1|\mathbf{y}] \approx \sum_{w_1, w_2} (w_1 - \mathbb{E}[w_1|\mathbf{y}])^2 q(w_1, w_2) \approx 0.01 \quad (7)$$

Hence, the posterior standard deviation of w_1 is $\sqrt{\mathbb{V}[w_1|\mathbf{y}]} \approx 0.07$

```
In [6]: # list all pairs (w1, w2) of non-zero prob
weights_and_probs = np.array([
    (0.5, 0.2, 0.12),
    (0.5, 0.3, 0.09),
    (0.6, 0.1, 0.12),
    (0.6, 0.2, 0.20),
    (0.6, 0.3, 0.12),
    (0.7, 0.1, 0.23),
    (0.7, 0.2, 0.12)])

mean_w1 = sum([w1 * p for (w1, w2, p) in weights_and_probs])
var_w1 = sum([(w1 - mean_w1)**2 * p for (w1, w2, p) in weights_and_probs])

print(f'Mean of w1: {mean_w1:3.2f}')
print(f'Variance of w1: {var_w1:3.2f}')
```

Mean of w1: 0.61
Variance of w1: 0.01

Question 2.2

The approximate posterior predictive distribution is

$$p(y^*|\mathbf{y}, x^*) = \int p(y^*|x^*, \mathbf{w})p(\mathbf{w}|\mathbf{y})d\mathbf{w} \approx \sum_{w_1, w_2} p(y^*|x^*, \mathbf{w})q(w_1, w_2) = \sum_{w_1, w_2} \mathcal{N}\left(y^*|e^{w_1+w_2x^*}, \frac{1}{2}\right) q(w_1, w_2). \quad (8)$$

Hence, the approximate posterior predictive distribution is a Gaussian mixture model with 7 components. We get

$$p(y^* = 4|\mathbf{y}, x^*) \approx \sum_{w_1, w_2} \mathcal{N}\left(y^* = 4|e^{w_1+w_2x^*}, \frac{1}{2}\right) q(w_1, w_2) \approx 0.29 \quad (9)$$

```
In [7]: npdf = lambda x, m, v: jnp.exp(-(x-m)**2/(2*v))/jnp.sqrt(2*jnp.pi*v)
f = lambda x, w1, w2: jnp.exp(w1 + w2*x)
y_star = 4
x_star = 4
sigma2 = 0.5

post_pred = sum([(npdf(y_star, f(x_star, w1, w2), sigma2))*p for (w1, w2, p) in weights_and_probs])

print(f'log p(y*=4|y, x*) is approximately {post_pred:3.2f}')
```

log p(y*=4|y, x*) is approximately 0.29

Part 3

The joint distribution is given by

$$p(y_1, z_1, \boldsymbol{\theta}) = p(y_1|\boldsymbol{\theta}, z_1)p(z_1)p(\boldsymbol{\theta}) \quad (10)$$

$$= [(1 - z_1)\mathcal{N}(y_n|0, 1) + z_n\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)] \text{Ber}(z_1|\pi)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) \quad (11)$$

Question 3.1

The marginal distribution is obtained marginalizing out z_1 using the sum rule

$$p(y_1, \boldsymbol{\theta}) = \sum_{z_1 \in \{0,1\}} [(1 - z_1)\mathcal{N}(y_n|0, 1) + z_n\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)] \text{Ber}(z_1|\pi)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) \quad (12)$$

$$= \mathcal{N}(y_n|0, 1)(1 - \pi)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) + \mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)\pi\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) \quad (13)$$

$$= (1 - \pi)\mathcal{N}(y_n|0, 1)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) + \pi\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) \quad (14)$$

Question 3.2

The marginal distribution is obtained marginalizing out $\boldsymbol{\theta}$ using the sum rule

$$p(y_1, z_1) = \int [(1 - z_1)\mathcal{N}(y_n|0, 1) + z_n\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)] \text{Ber}(z_1|\pi)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I})d\boldsymbol{\theta} \quad (15)$$

$$= \int [(1 - z_1)\mathcal{N}(y_n|0, 1)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I}) + z_n\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I})] \text{Ber}(z_1|\pi)d\boldsymbol{\theta} \quad (16)$$

$$= \left[(1 - z_1)\mathcal{N}(y_n|0, 1) \int \mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I})d\boldsymbol{\theta} + z_n \int \mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2)\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I})d\boldsymbol{\theta} \right] \text{Ber}(z_1|\pi) \quad (17)$$

$$= [(1 - z_1)\mathcal{N}(y_n|0, 1) + z_n\mathcal{N}(y_n|0, \sigma^2 + \alpha^{-1}\mathbf{x}_n^T \mathbf{x}_n)] \text{Ber}(z_1|\pi) \quad (18)$$

Question 3.3

The entropy of mean-field distribution is given by

$$\mathcal{H}[q_1^*] = -\mathbb{E}_{q_1^*(\boldsymbol{\theta})} [\log q(\boldsymbol{\theta})] = \frac{1}{2} \sum_{i=1}^D \log(2e\pi v_i) = \log(2e\pi 6/5) \approx 3.02 \quad (19)$$

Question 3.4

We have $q_1^*(\theta_2) = \mathcal{N}(\theta_2|1/4, 6/5)$ and hence, a 95% interval for θ_2 is

$$\left[\frac{1}{4} - 1.96\sqrt{\frac{6}{5}}, \frac{1}{4} + 1.96\sqrt{\frac{6}{5}} \right] \approx [-1.90, 2.40].$$

```
In [8]: m2, v2 = 1/4, 6/5
lower = m2 - 1.96*jnp.sqrt(v2)
upper = m2 + 1.96*jnp.sqrt(v2)
print(f'95% interval: [{m2 - 1.96*jnp.sqrt(v2):3.2f}, {m2 + 1.96*jnp.sqrt(v2):3.2f}]')
```

95% interval: [-1.90, 2.40]

Question 3.5

Let

$$\mathbf{m} = \frac{1}{4} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{V} = \frac{6}{5} \mathbf{I} \quad (20)$$

then the (approximate) posterior predictive distribution is given by

$$p(y^*|y_1, \mathbf{x}_*) = \int p(y^*|\boldsymbol{\theta}, \mathbf{x}_*)p(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta} \approx \int p(y^*|\boldsymbol{\theta}, \mathbf{x}_*)q(\boldsymbol{\theta})d\boldsymbol{\theta} \quad (21)$$

$$= \int [(1 - \pi)\mathcal{N}(y^*|0, 1) + \pi\mathcal{N}(y^*|\boldsymbol{\theta}^T \mathbf{x}_*, \sigma^2)] \mathcal{N}(\boldsymbol{\theta}|\mathbf{m}, \mathbf{V})d\boldsymbol{\theta} \quad (22)$$

$$= (1 - \pi)\mathcal{N}(y^*|0, 1) \int \mathcal{N}(\boldsymbol{\theta}|\mathbf{m}, \mathbf{V})d\boldsymbol{\theta} + \pi \int \mathcal{N}(y^*|\boldsymbol{\theta}^T \mathbf{x}_*, \sigma^2)\mathcal{N}(\boldsymbol{\theta}|\mathbf{m}, \mathbf{V})d\boldsymbol{\theta} \quad (23)$$

$$= (1 - \pi)\mathcal{N}(y^*|0, 1) + \pi\mathcal{N}(y^*|\mathbf{m}^T \mathbf{x}_*, \sigma^2 + \mathbf{x}_* \mathbf{V} \mathbf{x}_*^T) \quad (24)$$

$$= \frac{1}{5}\mathcal{N}(y^*|0, 1) + \frac{4}{5}\mathcal{N}(y^*|1/2, 1/10 + 12) \quad (25)$$

$$= \frac{1}{5}\mathcal{N}(y^*|0, 1) + \frac{4}{5}\mathcal{N}(y^*|1/2, 121/10) \quad (26)$$

Question 3.6

Since every mean-field Gaussian distribution can also be written as a full-rank Gaussian (i.e. with diagonal covariance matrix), we have that $\mathcal{Q}_1 \subset \mathcal{Q}_2$. Therefore, we have that

$$\text{KL} [q_1^*(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathbf{y}_1)] \geq \text{KL} [q_2^*(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathbf{y}_1)] \quad (27)$$

because we are optimizing over a larger family.

Part 4

Question 4.1

We have $\mathbb{V}(f(x^*)) = k(x^*, x^*) = 5(1 + 1) = 10$, and therefore,

$$p(f^*) = \mathcal{N}(f^*|0, 10).$$

Question 4.2

The prior distribution for $f(x^*)$ is zero-mean Gaussian, so by symmetry, the prior predictive probability for $y^* = 0$ for $x^* = 4$ is $p(y^* = 0|x^* = 4) = 0.5$. The same conclusion can be reached using the probit approximation or a Monte Carlo approximation.

Question 4.3

We have

$$p(f^*|\mathbf{y}, \mathbf{x}^*) \approx \mathcal{N}(f^*|\mu_{f^*|\mathbf{y}}, \sigma_{f^*|\mathbf{y}}^2) = \mathcal{N}(f^*|-0.71, 4.16),$$

for

$$\mu_{f^*|y} = \mathbf{k}^T \mathbf{K}^{-1} \mathbf{m},$$

$$\sigma_{f^*|y}^2 = k(x^*, x^*) - \mathbf{k}^T \mathbf{K}^{-1} (\mathbf{K} - \mathbf{S}) \mathbf{K}^{-1} \mathbf{k},$$

where $\mathbf{k}$ is the covariance between $f(x^*)$ and $\mathbf{f}$ and $\mathbf{K}$ is the prior covariance for $\mathbf{f}$ computed using k_1 .

```
In [9]: x = jnp.array([1, 2, 3])[:, None]
y = jnp.array([1, 1, 0])[:, None]
x_star = jnp.array([4])[:, None]

m = jnp.array([1.56, 0.84, -0.32])
S = jnp.array([ [2.69, 1.46, 0.42],
                [1.46, 1.78, 1.2],
                [0.42, 1.2, 2.13]
                ])

# use kernel implementation from week 6
from exercise6 import StationaryIsotropicKernel
kernel_fun = lambda tau, kappa=1, lengthscale=1: 5 + 5*jnp.exp(-tau**2/4)

# kernel function
kernel = StationaryIsotropicKernel(kernel_fun)
k = kernel.construct_kernel(x, x_star)
K = kernel.construct_kernel(x, x)

# compute mean and variance using equations from GP classification notebook
mu = k.T@jnp.linalg.solve(K, m)
v = jnp.linalg.solve(K, k)
sigma2 = kernel_fun(0) - v.T @ (K - S) @ v

# print
print(f'mean: {mu[0]:3.2f}')
print(f'var: {sigma2[0,0]:3.2f}')
```

```
mean: -0.71
var: 4.16
```

Question 4.4

The probit approximation yields

$$p(y^*|y, x^*) \approx \Phi \left(\frac{\mu_{f^*|y}}{\sqrt{\frac{8}{\pi} + \sigma_{f^*|y}^2}} \right) \approx 0.39$$

```
In [10]: from scipy.stats import norm
probit = lambda x: norm.cdf(x)
p_y_star = probit(mu/jnp.sqrt(8/jnp.pi + sigma2))[0,0]
print(f'The post. pred prob is {p_y_star:3.2f}')
```

```
The post. pred prob is 0.39
```

```
In [11]: mu = -1
sigma2 = 11
p_y_star = probit(mu/jnp.sqrt(8/jnp.pi + sigma2))
print(f'The post. pred prob is {p_y_star:3.2f}')
```

The post. pred prob is 0.39

Question 4.5

The kernel function controls the prior of the function and since the prior variance of $p(f(x))$ is $k(x, x)$. Therefore, the prior variance when using k_2 is 10 times smaller than for k_1 . Hence, the prior is more concentrated around zero, and therefore we expect the predictions to become more similar to the prior prediction. Therefore, we expect the posterior predictive probability for $y^* = 1$ to move closer to 0.5 when switching from k_1 to k_2 .

Part 5

Question 5.1

```
In [12]: # import metropolis sampler from course
from exercise9 import metropolis
log_npdf = lambda x, m, v: -0.5*jnp.log(2*jnp.pi*v) - (x-m)**2/(2*v)

def log_target(theta):
    z, x = theta
    return log_npdf(z, 1, 1) + log_npdf(x, 0, jnp.exp(z))

samples1, accept1 = metropolis(log_target, num_params=2, tau=jnp.sqrt(2), num_iter=3000, theta_init=jnp.array([-1, 0]))
samples2, accept2 = metropolis(log_target, num_params=2, tau=jnp.sqrt(2), num_iter=3000, theta_init=jnp.array([2, 5]))

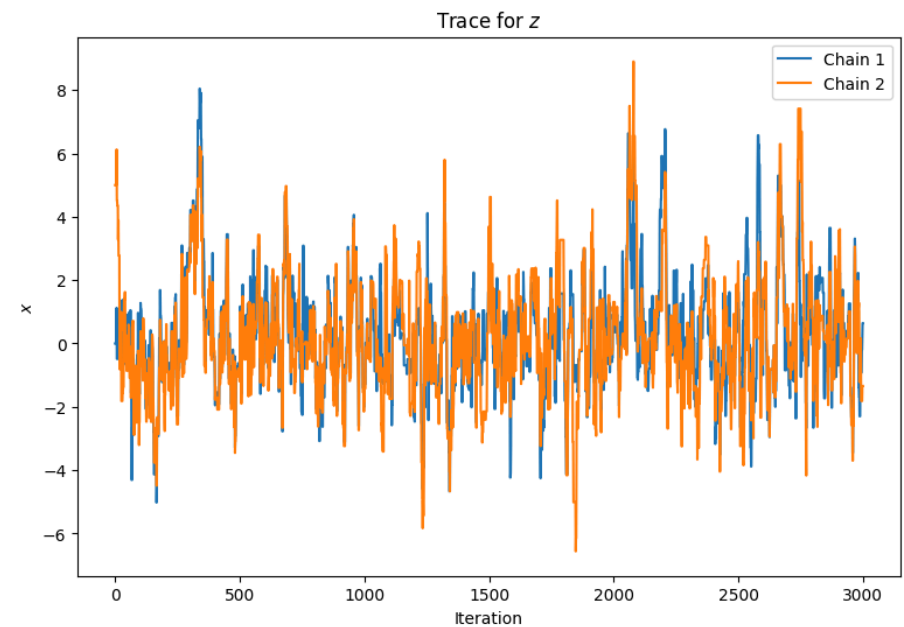
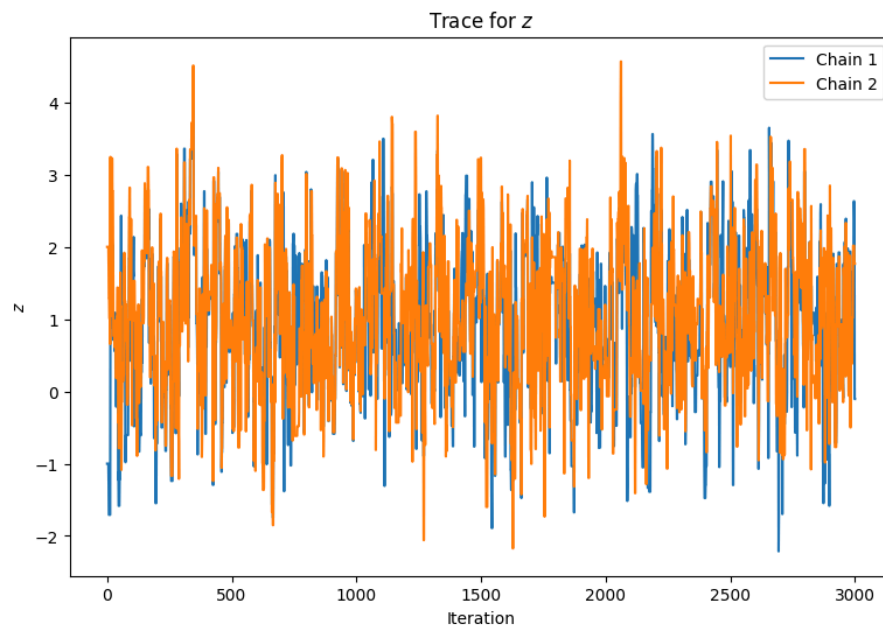
fig, ax = plt.subplots(1, 2, figsize=(20, 6))
ax[0].plot(samples1[:, 0], label='Chain 1')
ax[0].plot(samples2[:, 0], label='Chain 2')
ax[0].legend()
ax[0].set(title='Trace for $z$', xlabel='Iteration', ylabel='$z$')

ax[1].plot(samples1[:, 1], label='Chain 1')
ax[1].plot(samples2[:, 1], label='Chain 2')
ax[1].set(title='Trace for $x$', xlabel='Iteration', ylabel='$x$')
ax[1].legend()
```

Acceptance ratio: 0.46

Acceptance ratio: 0.47

Out[12]: <matplotlib.legend.Legend at 0x7f609ebec470>



Question 5.2

The mean of $f(x, z) = (z - x)^2$ is given by

$$\mathbb{E}[(z - x)^2] \approx \frac{1}{S} \sum_{i=1}^S (z^{(i)} - x^{(i)})^2 \approx 4.48 \quad (28)$$

```
In [13]: samples = jnp.concatenate([samples1, samples2])
z, x = samples.T
f = lambda theta: (z - x)**2
print(f'Mean is {jnp.mean(f(samples)):3.2f}')
```

Mean is 4.25

Question 5.3

The probability of the event $x < 1$ is

$$p(x < 1) = \mathbb{E}[x < 1] \approx \frac{1}{S} \sum_{i=1}^S \mathbb{I}[x^{(i)} < 1] \approx 0.66 \quad (29)$$

```
In [14]: f = lambda theta: 1.0*(x < 1)
print(f'Mean is {jnp.mean(f(samples)):3.2f}')
```

Mean is 0.73